SpeleoMesh

A digital communication system, designed for cave rescue

Dane Evans,
NSW Cave Rescue

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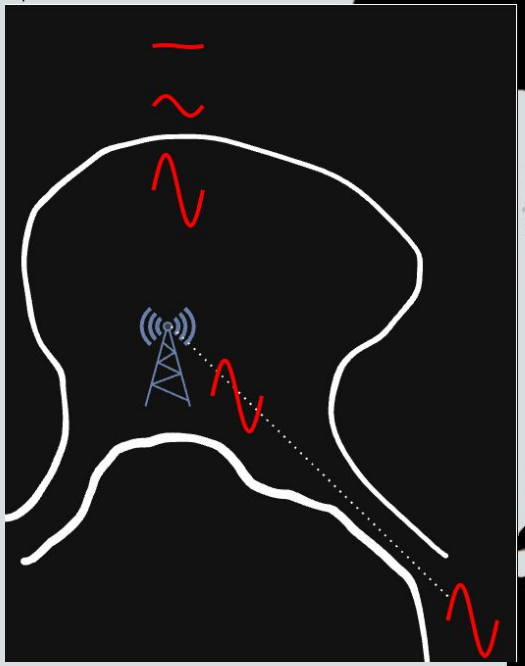
The Problems

Communication underground is inherently difficult.

Rock absorbs and/or reflects most radio signals. The variable nature of passages, chambers and squeezes make relying on this reflection impossible too.

Add to that the logistic issues - everything needs to be moved by hand through small passages, through water, and operate in place, potentially for several days.



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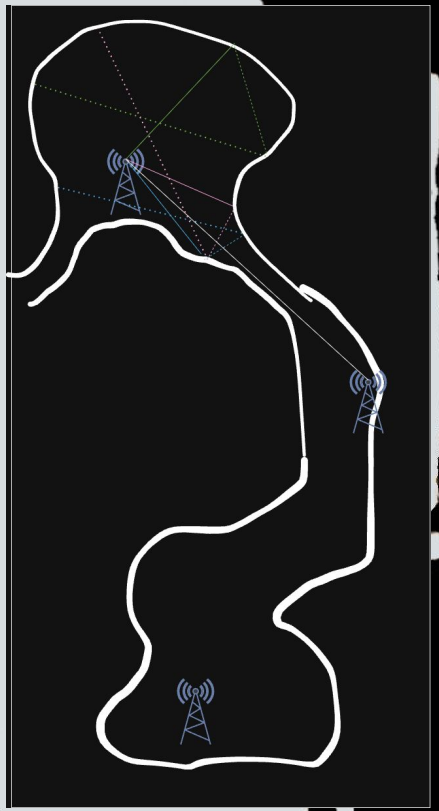

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Existing systems

Existing systems can be divided into 3 different groups

- Through rock radio
  - Cavelink / Nicola using low frequencies to transmit through the rock itself.
- Wired systems
  - Michie, Leaky feeder digital systems
- Adhoc radio systems
  - UHF / VHF line of sight systems

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SpeleoMesh | FLAMINGO

- Digital **mesh** system, based on open source Meshtastic®
  - Nodes will communicate to each other and rebroadcast messages
  - Self expanding and self healing as the physical positions of Nodes change.
- LoRa (**Lo**ng **Ra**nge) radio
  - 915MHz signal
  - Long range - I've seen up to 115km
  - Low power - up to a week on an 18650
- Low bandwidth,
  - Text
  - Telemetry (sensors / GPS)

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Advantages

- Text based has no issues noisy areas
- Small size, easy to carry
- Replaceable batteries, if you exceed a week
- Self contained
- Failures still allow localised comms either side of the break
- No tethering to a wire - you can be anywhere in sight of a trunk node
- Faster to lay than wires.
- Wired links are available for nasty sections
- Easily logged, or piped to an internet connection
- You get history, and can look back at messages.
- You can get messages in transit
- Reasonable prices, current hardware, and primarily digital means that making more in 10 years should be easy

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Disadvantages

- Care needs to be taken to ensure decent signal in both directions
- How many will be needed is variable.
- Signal is affected by human bodies in the path
- Tethered to a phone as an interface (currently)
- Data only, no voice
- Will need technical users to administer,at least in the short term.
- Phone / app stores pose some risks to long term use.

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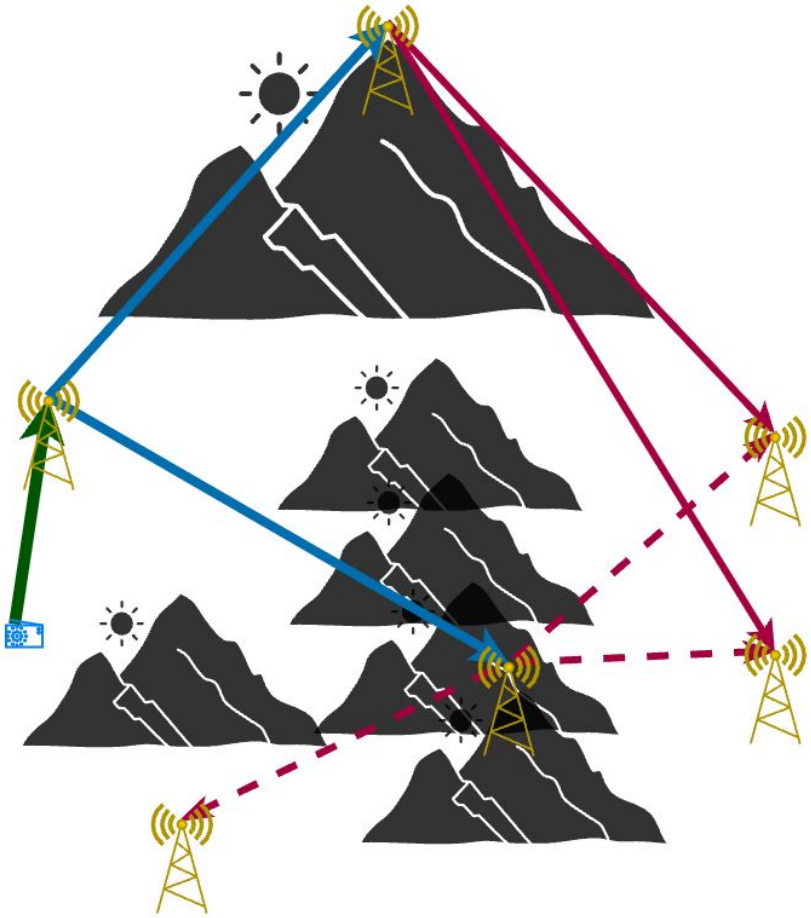
What’s a mesh

Above ground - Typical interconnected mesh

Our more linear mesh



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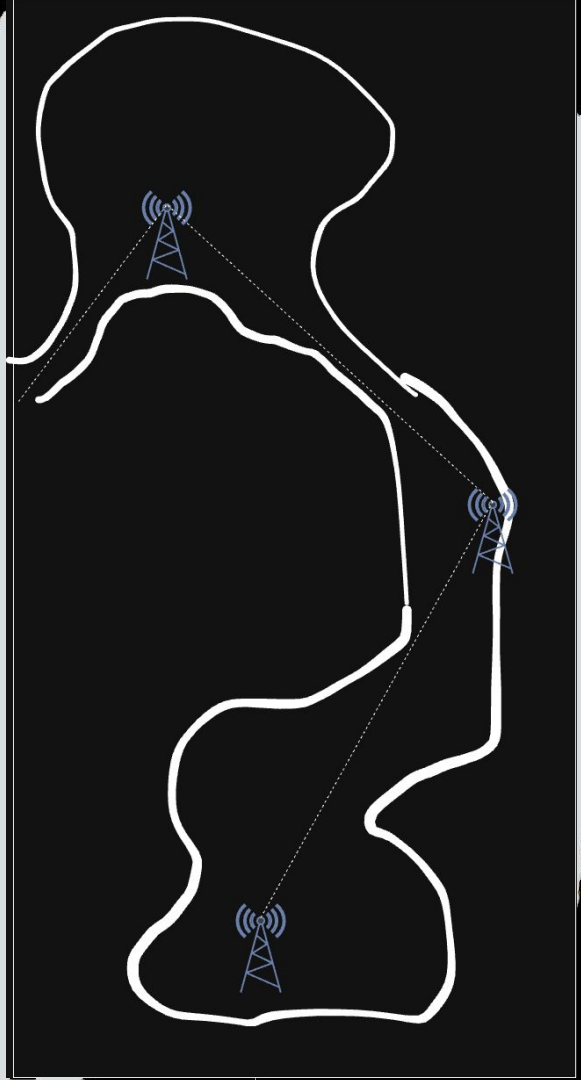

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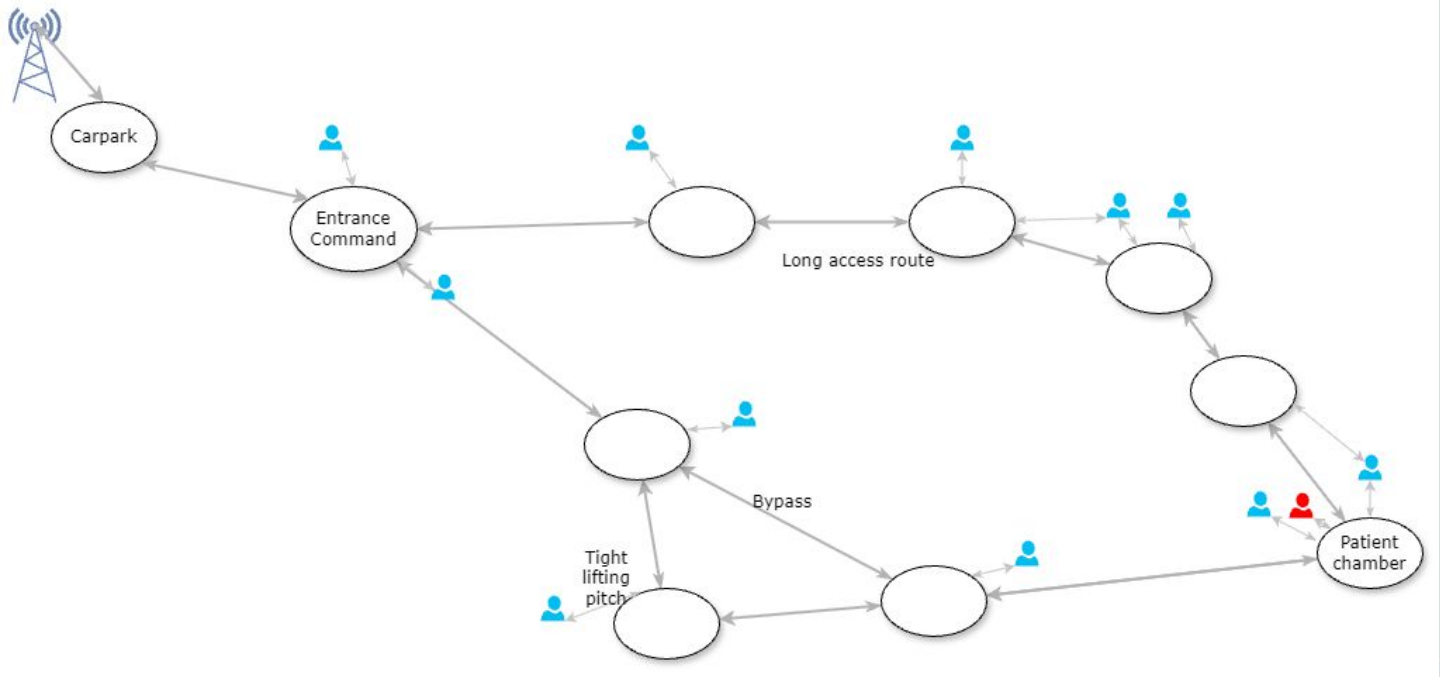
What's a mesh

Our topology leads itself to a 'Trunk & Personal Node' distinction.



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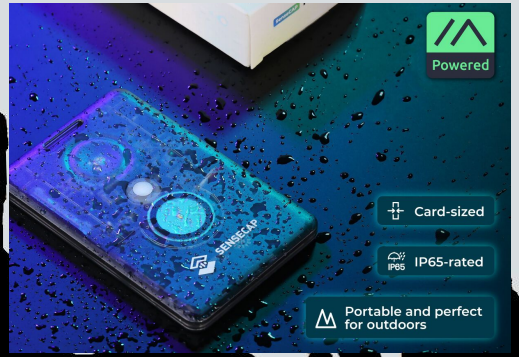
Hardware

While there are several commercial Meshtastic units available, most of them are *Development boards,* bare circuit boards, designed for testing and expansion.

The ones that come with cases are typically not water resistant, let alone waterproof, are power hungry, or just poorly designed for our uses.

There are a few that are built to a decent standard, and could be viable for use, especially as personal nodes.

- Seeed Studio T1000E - IP65, but small, 36 hour battery
- MuziWorks R1 Neo - “Water resistant”, and “multiday battery”



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Hardware

There are several devices that will be in use in the exercise

- CaveNode Basic Nodes
- CaveNode Basic w/ external antenna
- CaveNode Wired links

These are designed to be waterproof, and handle our environment, and have some additional LEDs to make placement in a reliable network easier.

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Getting on the network

To get onto the network, you just need a Node, and a phone.

The open source app is available on both app stores, and will talk to a node via bluetooth.

- configuration changes to be made to optimise cave use

Once you are on the network, your packets will be repeated by the mesh, but messages can be separated into encrypted channels.

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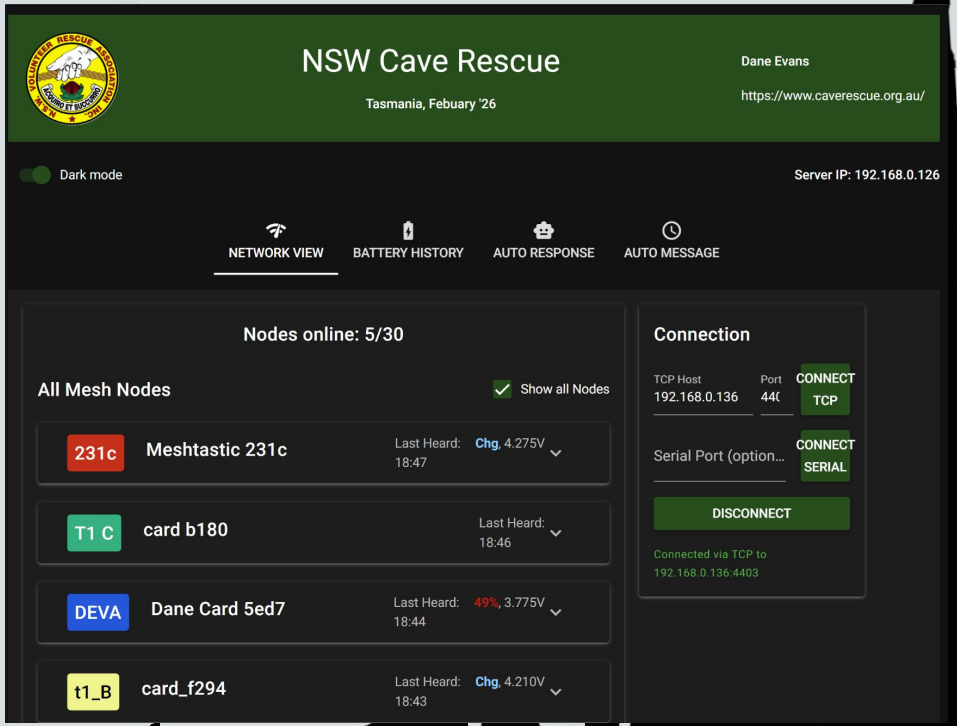
Command software

Live webapp, viewable on either laptop or mobile

- Network health:
  - See batteries, and spot issues in the network
- Automated responses:
  - Check that messages reach the surface
- Automated messages
  - Reminders
  - Set LED states
- Network Map (coming soon)
- Messaging(coming soon)



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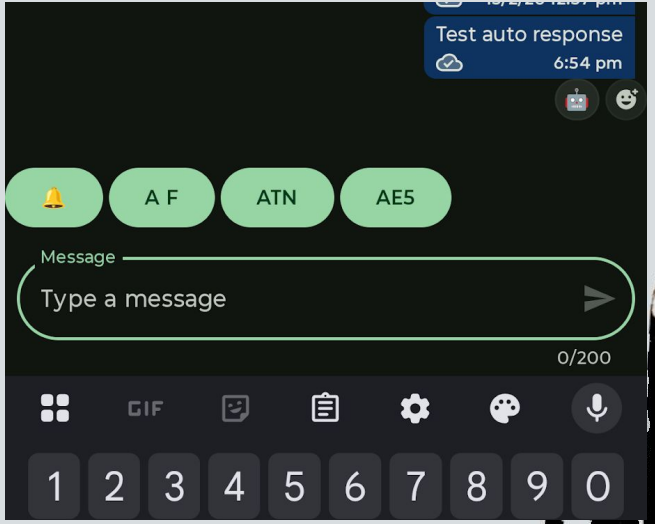

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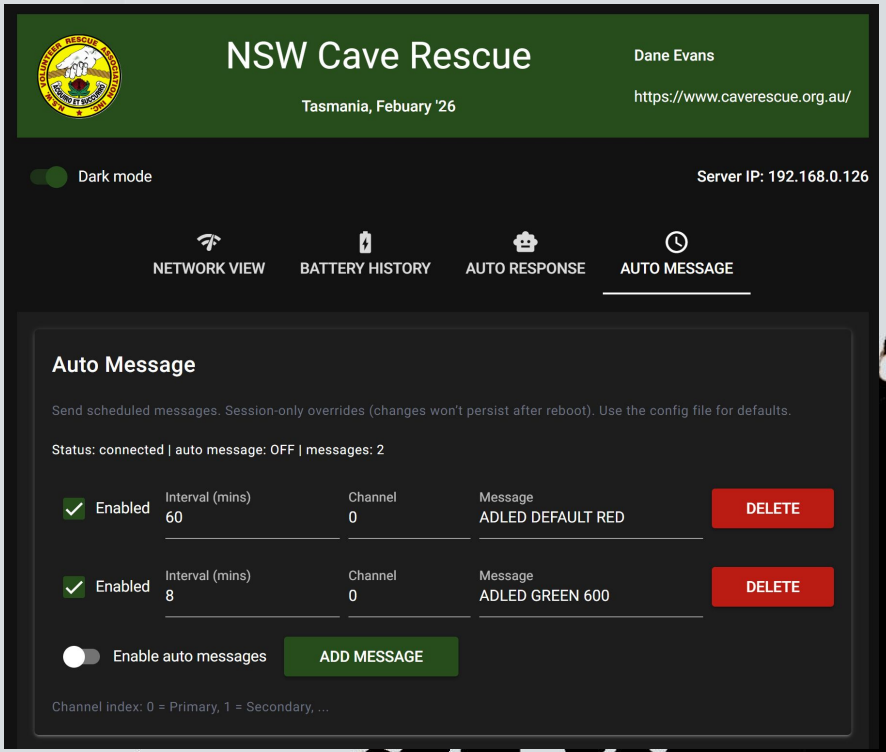

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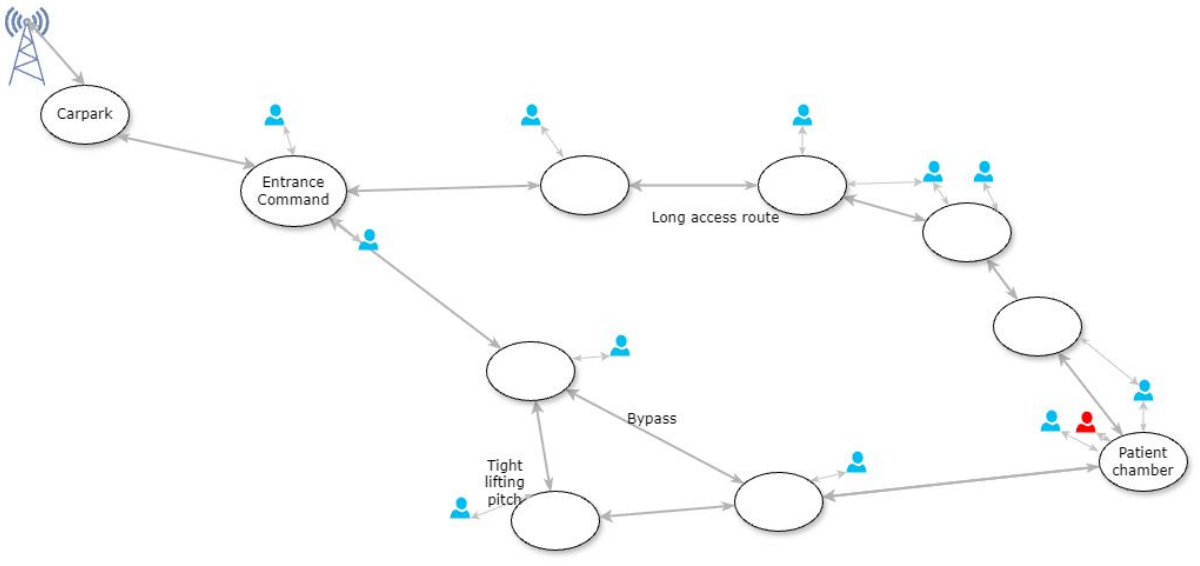

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Where to from here?

- Improvements to Surface control software
  - Send and receive messages,
  - Network mapping
- Logging to SDcard from a dedicated node
- Bridges to external systems - internet via mobile internet / starlink
- Telemetry - temperature and CO2 monitoring
- Cutdown ‘user’ app, to remove settings and other features we don’t use
- A stand alone node design, which will handle the conditions and eliminate the reliance on the whims of Google / Apple

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Further reading

SpeleoMesh project
Docs - https://daneevans.github.io/SpeleoMesh/

Cave Firmware
https://github.com/DaneEvans/Flamingo-Firmware

Meshtastic®
Docs - https://meshtastic.org/docs/introduction/
Repositories - https://github.com/meshtastic

Huntsville Cave Rescue Unit - Flamingo project (Alabama, USA)
https://github.com/rbreesems/flamingo

Vangelis - Cave Rescue Meshtastic project (Virginia, USA)
https://github.com/semper-ad-fundum/vangelis